

CASE STUDY

Match 3 Levels

Royal Match, Homescape,
Match Villains and Merge garden

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A quick glossary

before we move ahead...

- **Grid**: this refers to the matrix/level with the tiles
- **Breather Level** : is a level (or set of levels) intentionally placed to offer a change of pace. It avoids introducing new mechanics and instead focuses on reiterating previously learned ones

Games

This study examined **four** games, focusing on a comparative analysis of their level design, grid implementation, and element interaction.



Royal match



Homescape



Match Villains



Marge Gardens

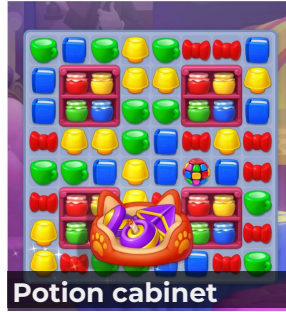
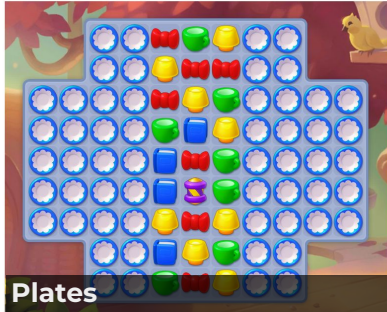
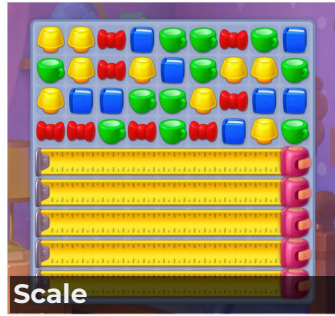
Design Patterns

identified across all Games

Introducing new mechanics

- **Foundation & Introduction:** New game mechanics are initially introduced on **simple, standard grids** (square or rectangle).
- Grid complexity is only increased when the mechanic requires specific structural emphasis.

(image on the left) A collage of grids from Homescape, used for introduction of new mechanics



1st level

★ Item introduction ★

2nd

Item used with **1 previously**
introduced mechanic

3rd

Item used with **1 or more**
previously introduced mechanic

4th

Item used with **1 or more**
previously introduced mechanic

5th

Item used with **1 or more**
previously introduced mechanic

Next levels

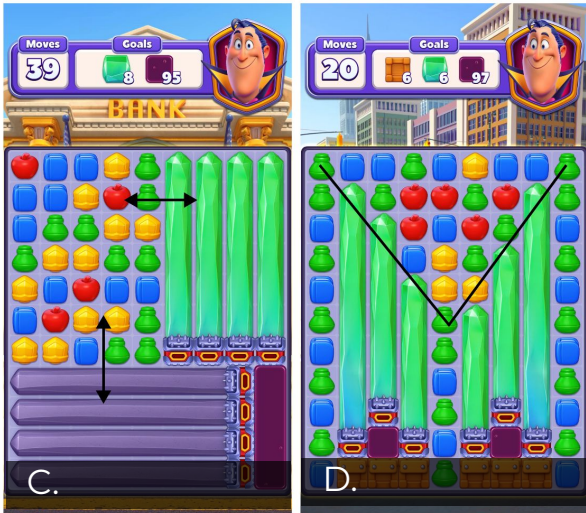
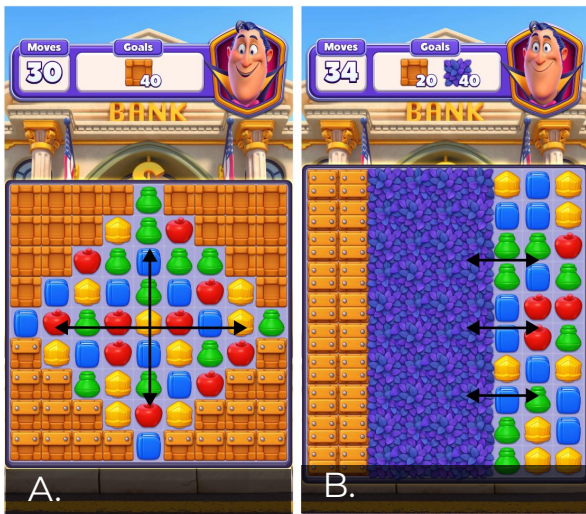
Design Patterns

identified across all Games

Introducing new mechanics

- **Pacing** is generally designed on a 10-20 level cycle: A major new mechanic or item is introduced at the start of each cycle (Level 1, 11, 21, etc.).
- Following its introduction, each new mechanic or item is used in the **next 4-5 levels** to show its behavior in combination with other mechanics.





Design Patterns

identified across all Games

Booster Creation

- Boosters & Guidance:** Level design is intentionally structured to encourage players to create the specific boosters needed for success. Grids often enforce **directional booster** usage to clear specific, targeted areas.

(image on the left) A collage of grids from Match villains, shows how levels encourages directional booster creation



Design Patterns

identified across all Games

Difficulty Design

- Challenge is progressively increased through targeted grid modifications, including:
 - Introducing **hard-to-reach** cells or restricted areas.
 - Mandating complex **sequential clears**.
 - **Strategic arrangement** and combination of previously introduced mechanics.
 - Usage of a mechanic or a combination of mechanics in the level

(image on the left) A collage of grids from Merge Gardens, shows how level difficulty is designed



Games in details





Royal Match

Analysis details:

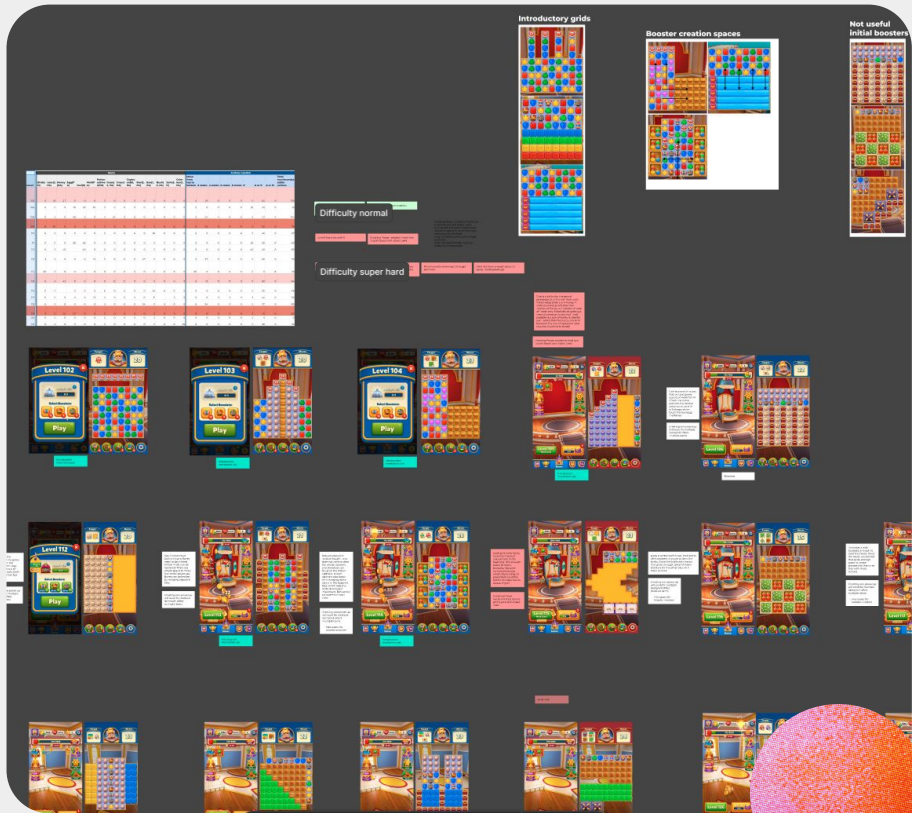
- Played on Phone
- Portrait mode
- Levels played: 100-160
- Core mechanic: Match 3
- Has tasks to improve different areas, players have no control how the area is designed.



Introducing mechanics

- **Introduction cycle:** A new game element is introduced every 20 levels
- **Introduction Level:** Level 1 focuses exclusively on the new element.
- **Integration:** The next 4–5 levels integrate the new element, exploring its interactions with the pre-existing mechanics.
- **Pacing:**
 - The new element is utilized in 10 of the 20 levels in the cycle.
 - Interspersed "breather" levels strategically combine previously introduced elements, offers variation and prevents mechanic fatigue.

Analysing Royal match, notes created on Figma
Table for mechanics used per level, on spreadsheet





Grids for introduction

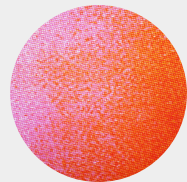
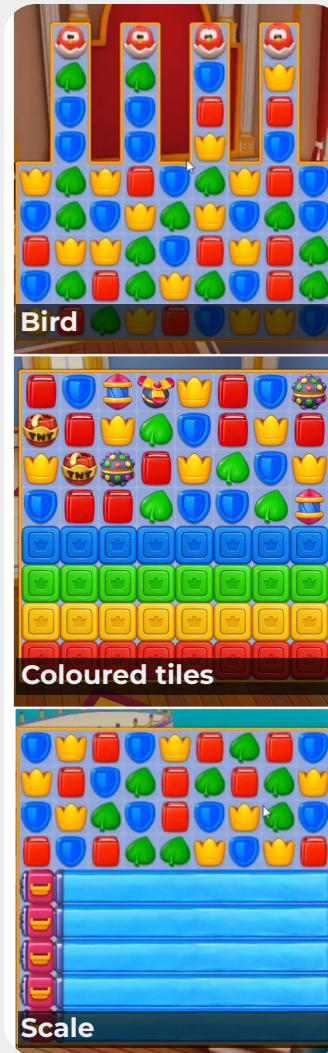
- **Use of square grids:**

Introductory mechanics are presented on 8x8 grids with variations tailored to the concept.

- **Grids modified for mechanics:**

Each grid's geometry is designed to visually reinforce the mechanic's core features.

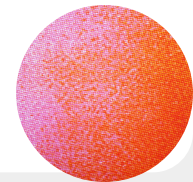
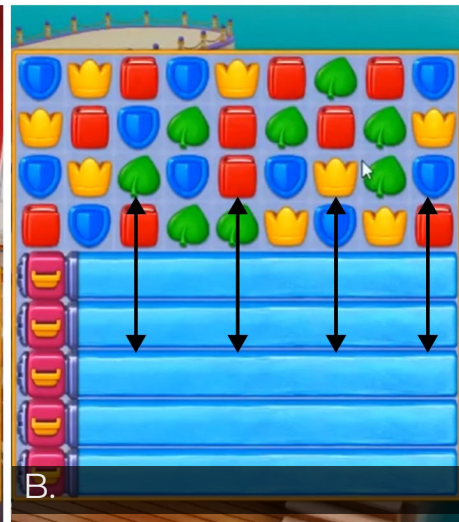
For example, the introduction of a downward-moving "Bird" mechanic utilizes a grid emphasizing vertical columns to naturally guide the eye and reinforce the required movement goal.





Spaces and Booster making

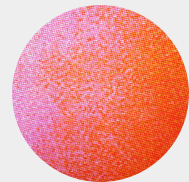
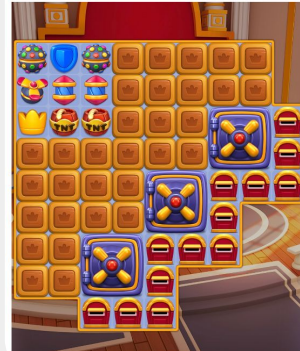
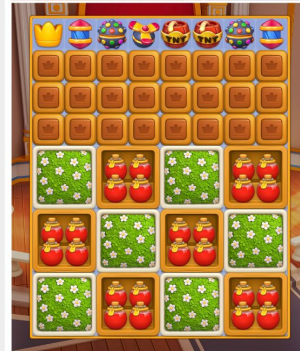
- **Bidirectional clearing:** Levels (such as level A) are designed to encourage bi-directional booster creation, rewarding players who look for both horizontal and vertical opportunities.
- **Vertical clearing:** Levels (such as level B) are structured to prioritize the creation of vertical clearing boosters.
- **Cross clearing:** Levels (such as level C) feature higher probability and optimal spacing to strongly encourage the player toward creating vertical, horizontal and cross-clearing boosters.





Boosters and Feel-good-situations

- **Initial boosters:** Levels are strategically seeded with initial boosters to provide an early advantage and expand the accessible play space.
- **Does boosters mean success?:** While the initial boosters offer a positive psychological incentive, they do not guarantee victory.
- **Cost of using boosters:** Players must spend turns to use these boosters, managing the trade-off between immediate impact and optimal tile pieces.





Designing difficulty

Difficulty Pacing: Challenge is intentionally paced, with significant difficulty spikes occurring every 5th level (**Hard**) and 9th level (**Super Hard**) within a 10-level block

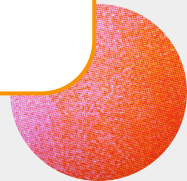
Sequential Clearing: Level progression often requires **sequential action**; critical areas of the level are only accessible or *breakable* after clearing prerequisite sections.

Targeted Constraints: Levels restrict player action, forcing players to:

- Execute **precise, targeted actions** (boosters/power-ups) on specific rows, columns, or cells.
- Work with a **limited move count**, increasing the pressure on efficient action.

Booster Bottleneck: The primary source of difficulty stems from **grid constraint** and object placement, which actively limits the player's ability to:

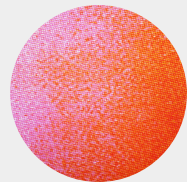
- **Generate necessary boosters** due to insufficient open space.
- Create **multiple boosters** simultaneously in advantageous positions (same row/column).
- The grid layout makes **power-up creation** difficult, often necessitating specific power-ups to target individual rows or isolated areas.





Difficulty creation: With Grids

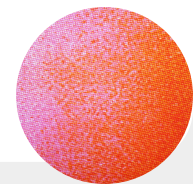
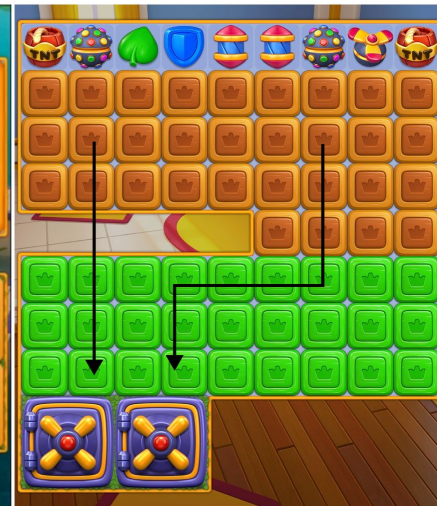
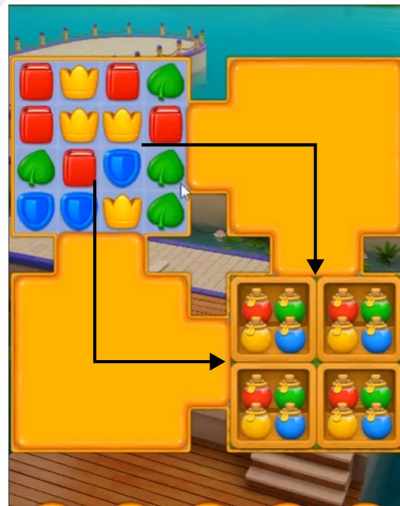
Difficulty is achieved by restricting booster access. Grid design ensures that not every cell is reachable by every available power-up.

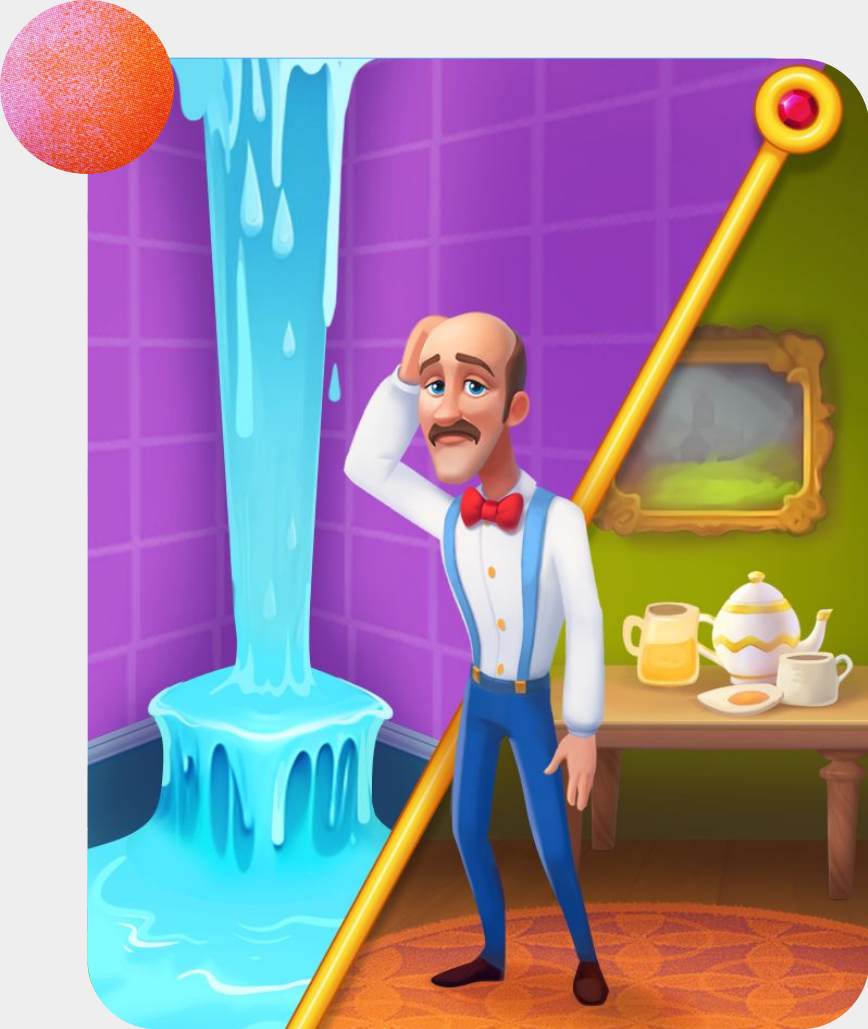




Difficulty creation: Areas within Grids

- **Sequential Access:** Critical areas of the grid are locked or gated, only becoming accessible after a preceding, required section is cleared.
- **Forced Prioritization:** This design forces players to strategically prioritize their clearing targets and consequently the specific boosters needed to unlock the path forward.





Homescape

Analysis details:

- Played on Tablet
- Landscape mode
- Levels played: 1-60
- Core mechanic: Match 3
- Has a narrative as main element of the game
- Has story, players have control in designing the home.



- [illegible]





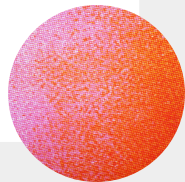
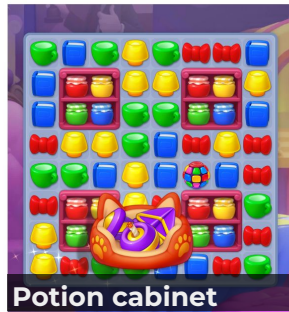
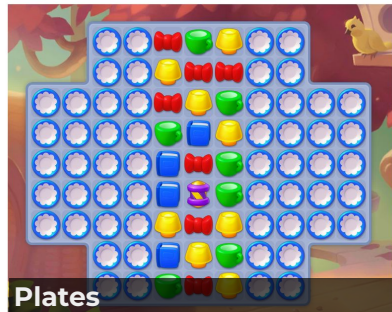
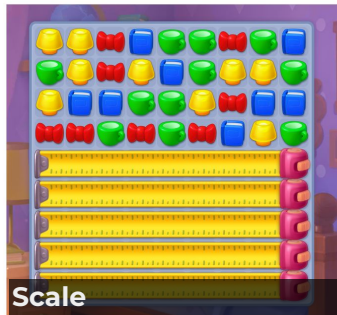
Grids for introduction

- **Use of Square grids:**

Grids used to introduce a mechanic are **generally 9x9** or **8x8** grids, with minor variations based on the mechanic introduced.

- **Grids modified for mechanics:**

For instance, a Plate breaks if tiles in the adjacent cells match. The variation of grid used for Plates, works for the introduction as this shape allows maximum breaking of plates, with enough space to make horizontal boosters.

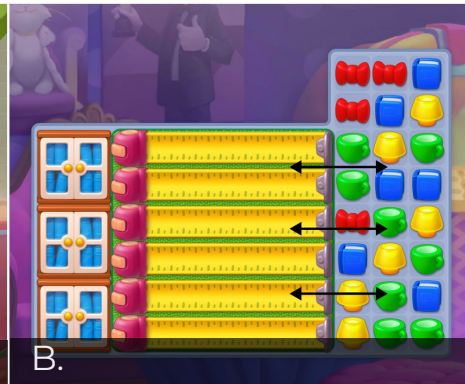
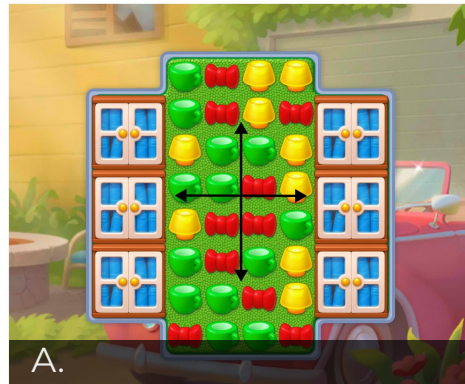




Spaces and Booster making

Level geometry is strategically arranged to incentivize the creation of specific boosters necessary for level completion.

- **Cross clearing:** levels (such as level A) are designed to encourage bi-directional booster creation, rewarding players who look for both horizontal and vertical clearings.
- **Horizontal clearing:** Levels (such as level B) are built to prioritize the creation of horizontal clearing boosters.
- **Bidirectional clearing:** Levels (such as level C) encourage the player toward creating vertical and cross-clearing boosters.
- **V-shaped arrangements:** Elements in levels are arranged in V shapes, to facilitate tiles to seep into gaps. This in turn allows players to create boosters, and reach different areas of the grid.

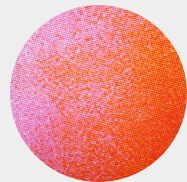




Boosters and Feel-good-situations

Similar to Royal match,

- **Initial boosters:** Levels are strategically seeded with initial boosters to provide an early advantage and expand the accessible play space.
- **Does boosters mean success?:** While the initial boosters offer a positive psychological incentive, they do not guarantee victory.
- **Cost of using boosters:** Players must spend turns to using these boosters, managing the trade-off between immediate impact and optimal tile pieces.





Designing difficulty

Pacing Spikes: Difficulty follows a calculated rhythm, with major spikes engineered at levels 3rd, 5th and 9th within every 10-level set

Sequential Objectives: Grid progression is often gated. Players must clear specific prerequisite areas to unlock access to the main objective areas.

Precision Demanded:

Success requires targeted action. Players must generate specific boosters or take actions that hit precise rows, columns, or cells.

Move & Space Economy:

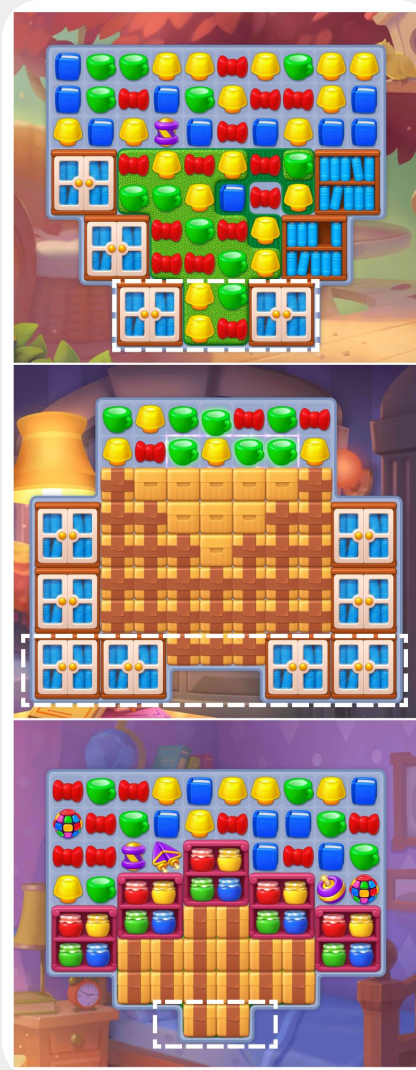
Challenge is magnified by simultaneous constraints:

- Limited moves for targeted action.
- Insufficient space to easily create required boosters.
- Grid design actively prevents multiple boosters in critical clear zones.



Difficulty creation: With Grids

Restricted Access: Grid layouts are designed to ensure not every cell is reachable by every booster type.

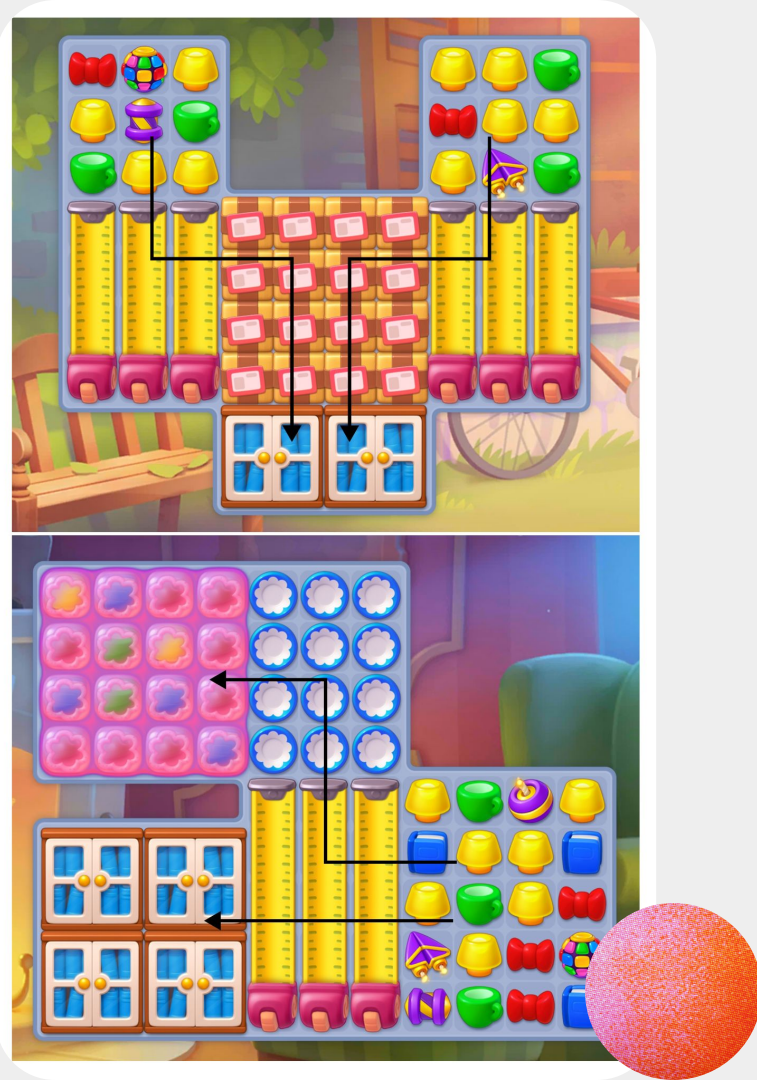




Difficulty creation: Areas within Grids

Unlock-to-Reach: Clearing one area is required to access the next.

Prioritize Boosters: Players must decide and create boosters based on this clearing sequence.





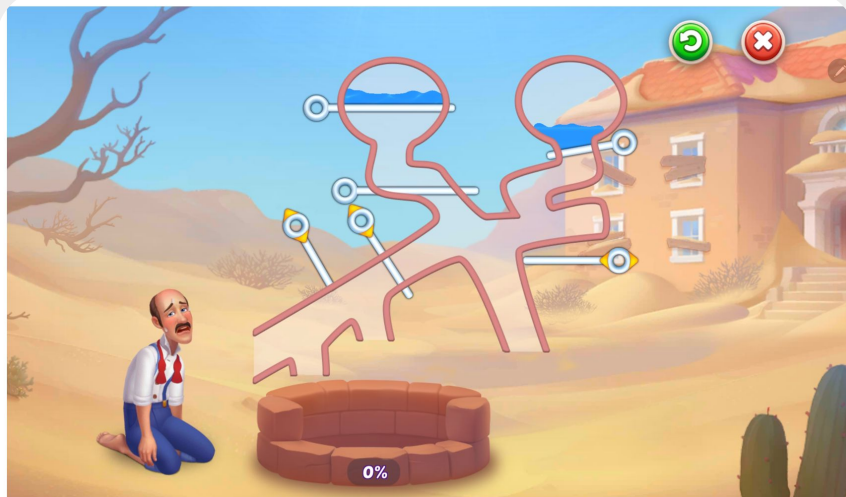
Minigames

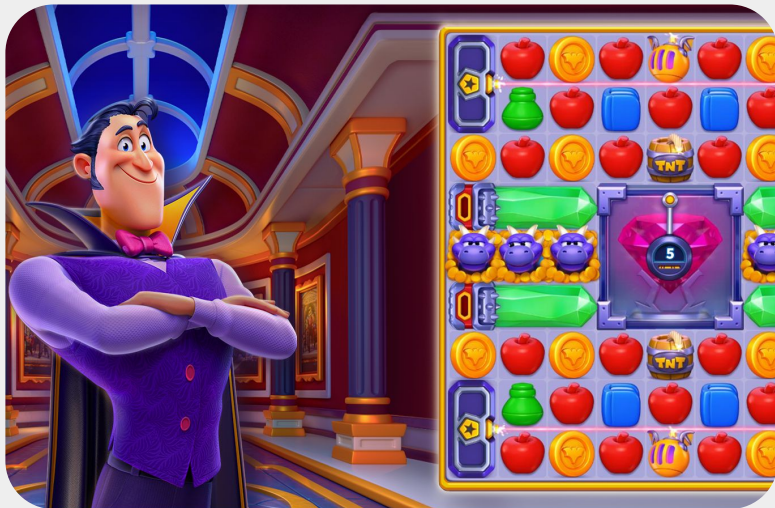
Pacing Variety: A puzzle level is strategically introduced after every two match-3 levels.

Player-Driven Tasks: Side tasks are non-linear and can be completed at the player's leisure, prioritizing player choice.

Dynamic Maps: Event maps are customizable, shifting based on player preference or mood.

Mini-Game Density: Within every 10 levels, 5 mini-games are included, typically appearing after every alternate level.





Match Villains

Analysis details:

- Played on phone
- Portrait mode
- Core mechanic: Match 3
- Levels played: 1-50
- Does not have a narrative as main element of the game
- USP: Heist mechanic



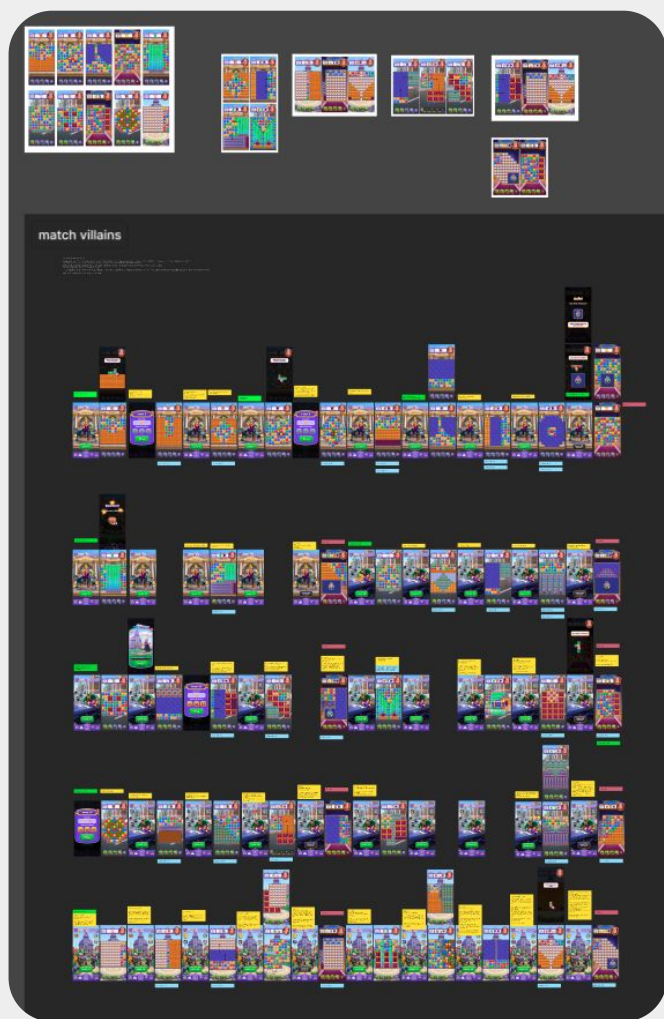
Introducing mechanics

Introduction Cycle: A new mechanic is introduced in every 10 levels within this section (Levels 21-50).

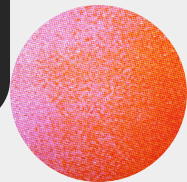
Introduction Level: 1st level serves as a tutorial level with regular tiles

Integration:

- The first three appearances of the new mechanic occur in non-sequential levels.
- The mechanic is revisited in the 9th and 10th levels of the cycle, before the next mechanic is introduced.



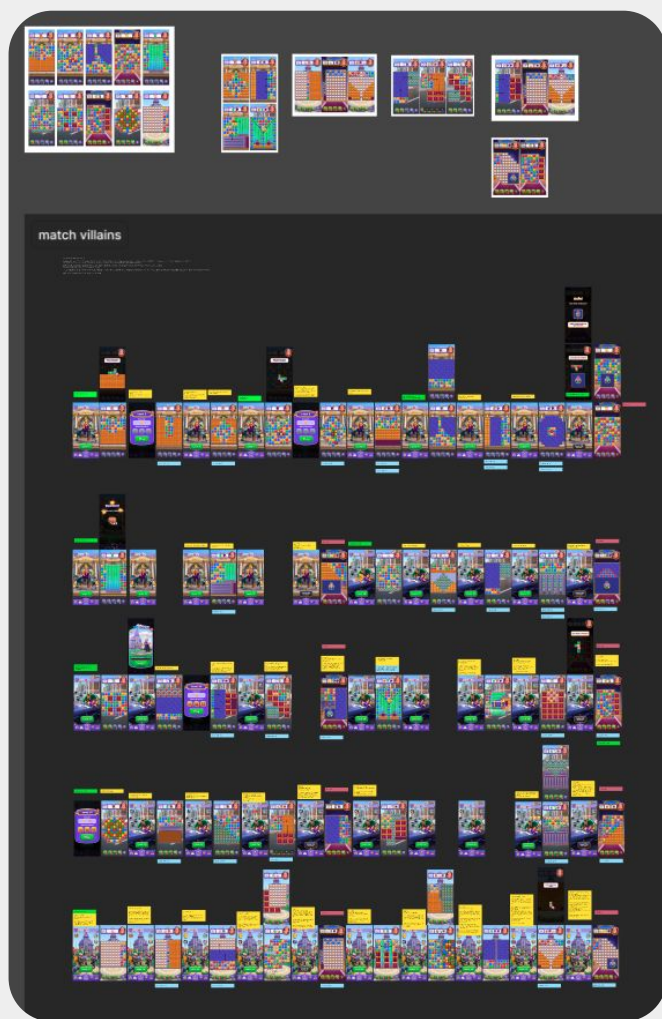
Analysing Match Villains, notes created on Figma



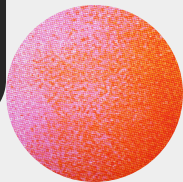


Introducing mechanics

- **Pacing:** Five 'breather levels' are strategically interspersed throughout each 10-level cycle. These levels combine previously introduced mechanics, offers variation and prevents player fatigue.
- **Heist Mechanic:** Heist mechanic is implemented in every 10th level. This marks the end of one cycle



Analysing Match Villains, notes created on Figma





Grids for introduction

- **Use of Square grids:** Grids used to introduce a mechanic were 9x9 grids, with minor variations.
- **Grids modified for mechanics:** For mechanics such as Ghost, Ice scale, laser can be broken only when their adjacent cells are merged - 9x9 grids were modified to allow easier affordances for these mechanics.
- **Layer-able Mechanic:** While a standard 9x9 is functional for cascading mechanics like Soil (which breaks from above), a diamond-shaped grid was strategically employed. This non-standard geometry increases the level tightness and complexity of downward flow, resulting in a more dynamic and engaging puzzle.

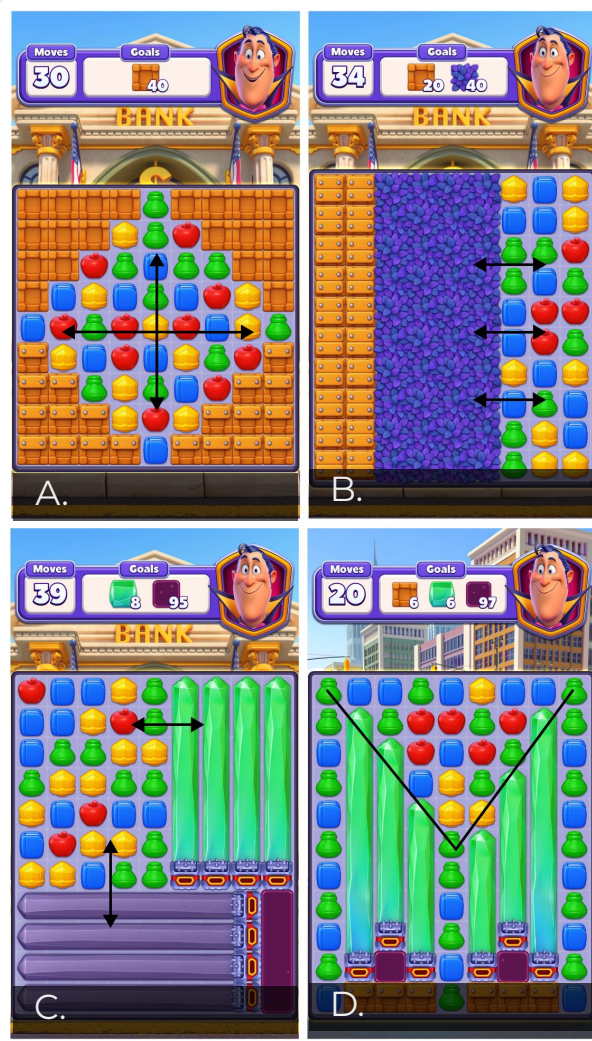




Spaces and Booster making

Booster placement in Match Villains **resembles** the design strategies used in *Royal match* and *Homescape*.

- **Cross clearing:** levels(such as level A) are designed to encourage bi-directional booster creation, rewarding players who look for both horizontal and vertical opportunities.
- **Horizontal clearing:** Levels(such as level B) are built to prioritize the creation of horizontal clearing boosters.
- **Bidirectional clearing:** Levels(such as level C) encourage the player toward creating vertical and cross-clearing boosters.
- **V-shaped arrangements:** Elements in levels are arranged in V shapes, to facilitate tiles to seep into gaps. This in turn allows players to create boosters, and reach different areas of the grid





Boosters and Feel-good-situations

Similar to Royal match and Homescape.

- **Initial boosters:** Levels are strategically seeded with initial boosters to provide an early advantage and expand the accessible play space.
- **Does boosters mean success?:** While the initial boosters offer a positive psychological incentive, they do not guarantee victory.
- **Cost of using boosters:** Players must spend valuable turns to deploying these boosters, managing the trade-off between immediate impact and optimal tile pieces.
- **Booster placement** within the level was **similar** to placements in Royal Match





Designing difficulty

Uses similar strategies with Royal Match and Homescapes

Pacing Spikes: Difficulty follows a calculated rhythm, hard levels placed in 5th and 10th level within every 10-level set

Sequential Objectives: Grid progression is often gated. Players must clear specific prerequisite areas to unlock access to the main objective areas.

Precision Demanded:

Success requires targeted action. Players must generate specific boosters or take actions that hit precise rows, columns, or cells.

Move & Space Economy:

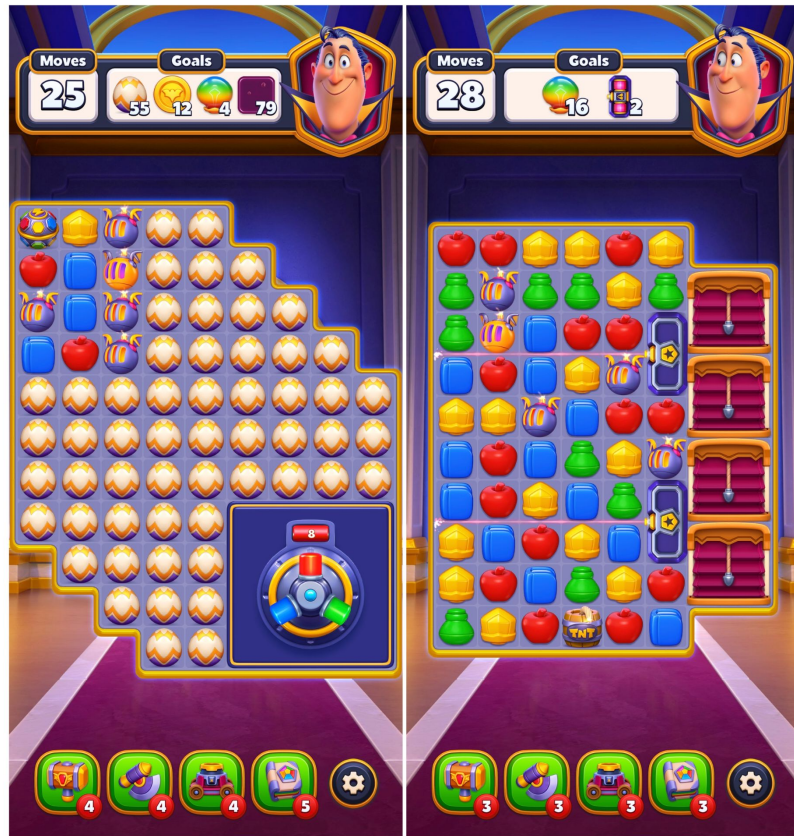
Challenge is magnified by simultaneous constraints:

- Limited moves for targeted action.
- Insufficient space to easily create required boosters.
- Grid design actively prevents multiple boosters in critical clear zones.



Difficulty creation: With Grids

- Level difficulty is designed by designing grid layouts that **restrict booster accessibility**, ensuring that specific cells are not reachable by every booster type.
- This requires players to mindfully plan booster creation and placement to successfully clear target areas.





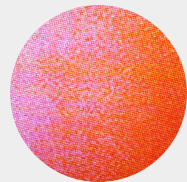
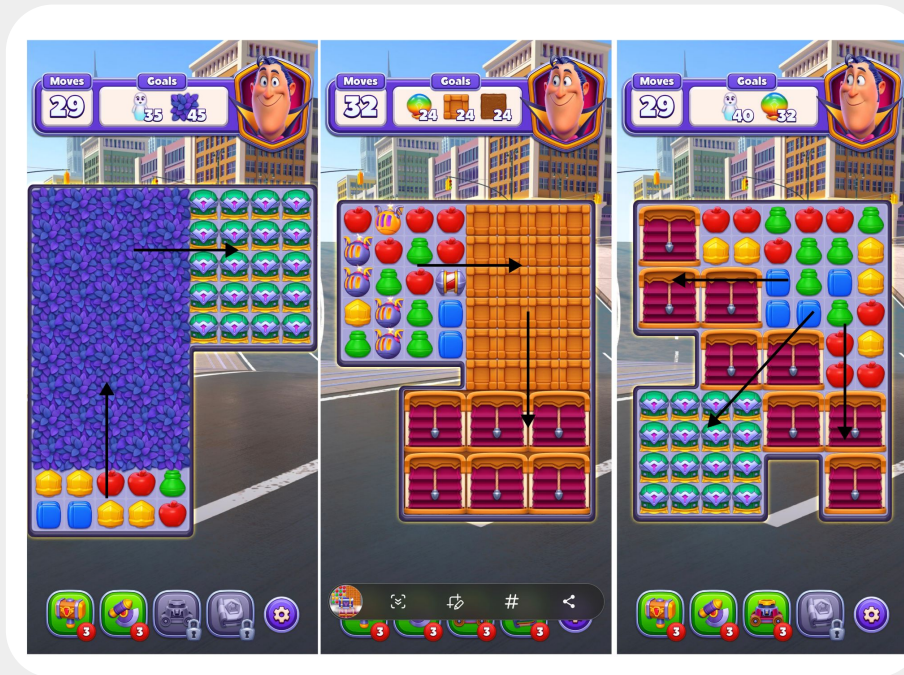
Difficulty creation: Areas within Grids

Level progression utilizes **sequential clearing**, where certain areas of the grid are only made accessible after an initial clearance has been achieved.

This deliberate blockage forces **players** to **prioritize their target zones** and create the boosters for an optimal opening move.

For example:

- In linear layouts (like the first and third levels), the optimal path is clear (e.g., clear Obstacle A to access Obstacle B).
- In non-linear layouts (like the third level), the player is given agency to choose their starting strategy.

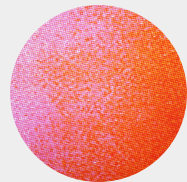




Difficulty creation: Jelly mechanic

Jelly levels, which requires full-grid propagation, are subject to unintentional difficulty spikes driven by the move count.

Additionally, the low visual prominence of the initial Jelly cells may often lead players to misallocate early moves - resulting in a steep difficulty curve.





Merge Gardens

Analysis details:

- Played on Phone
- Portrait mode
- Core mechanic: Merge 3
- Match 3 levels exists as puzzle levels to gather resources
- Levels played: 1-80
- Similar to Homescape, with merge mechanics
- Narrative acts as main the binding element of the game



Introducing mechanics

Introduction Cycle: A new mechanic is introduced every 20-25 levels within this section (Levels 20-80).

Introduction Level: 1st level serves as a tutorial level with regular tiles.

Between level 20-80, 3 new mechanics were introduced: first on 31st level, 2nd on 51st level, and 3rd on 76th level.

Integration: After each introduction, the next 5 levels had the mechanic used, and the next usage were randomized(at the moment it looks randomized)

Pacing: Once introduced, a new mechanic is used 11-16 times(11 times between level 31-50, and 16 times between level 51-75.).

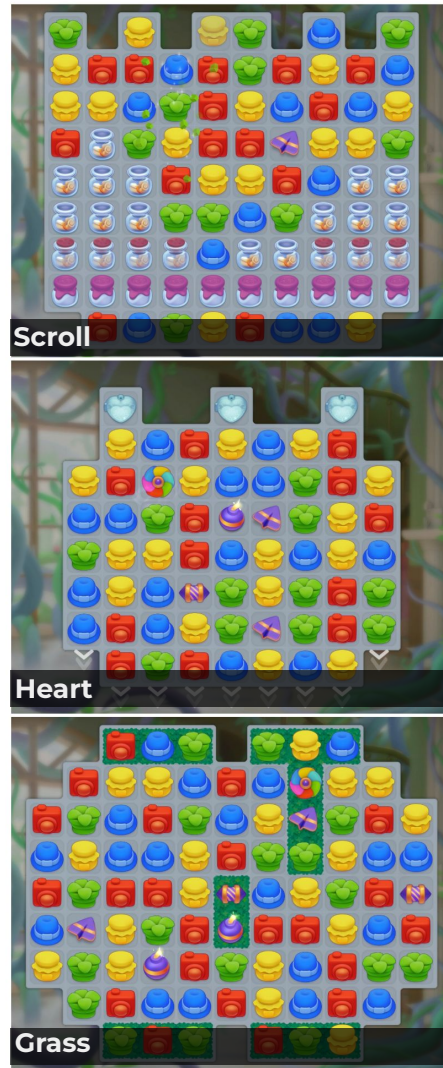


Analysing Merge Gardens, notes created on Figma



Grids for introduction

- **Grids modified to fit the mechanic introduced:** modified versions of Square grids and rectangular grids are used to introduce a mechanic.
- There was no constant pattern, rather it was based on the mechanic introduced, and arrangement of elements.
- **Use of multiple design decisions:** The level has a combination of grid modification, element placement and space for booster creation, that would allow easier clearing of the introduced elements





Spaces and Booster making

Depending on the required booster for the level, level items are arranged in a way to encourage booster creation.

- **Cross clearing:** Levels (such as level A) encourage players to create boosters in both the directions
- **Vertical clearing:** Levels (such as level B) encourage creation of vertical clearing boosters
- **Bidirectional clearing:** Levels (such as level C) encourage creation of both horizontal and vertical boosters, has space and probability of making vertical ones more
- **V-shaped arrangements:** Regular tiles are placed in V shape. As more Ice scale clears up, tiles can seep into gaps more, encouraging booster creation of different kinds



A.



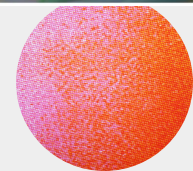
B.



C.



D.





Designing difficulty

Uses similar strategies with Royal Match, Homescapes and Match Villains

Pacing Spikes: Difficulty follows a calculated rhythm, with major spikes engineered at every 6th level.

Sequential Objectives: Grid progression is often gated. Players must clear specific prerequisite areas to unlock access to the main objective areas.

Precision Demanded:

Success requires targeted action. Players must generate specific boosters or take actions that hit precise rows, columns, or cells.

Element combinations:

Combinations of elements used in the level, their arrangement within the level and hits required to clear them had effects on the level of difficulty of the level.

Move & Space Economy:

Challenge is magnified by simultaneous constraints:

- Limited moves for targeted action.
- Insufficient space to easily create required boosters.
- Grid design actively prevents multiple boosters in critical clear zones.



Difficulty creation: With Grids

- Levels are design for sequential access, where key grid areas are initially gated and only become accessible after an adjacent section is cleared.
- This constraint immediately compels the player to strategically prioritize their initial target area and plan the appropriate booster creation for it.
- Suboptimal choices made may result in a loss of moves, increasing difficulty for players to complete the level.

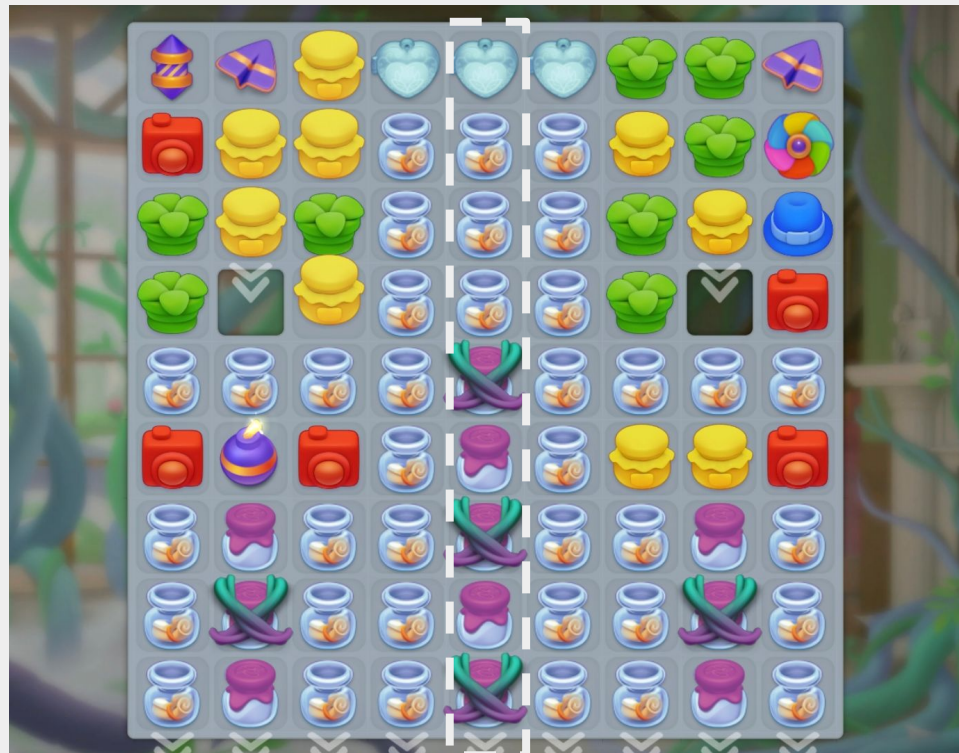




Difficulty creation: Combining elements

Difficulty is also designed by using different element, and their stages in **combination with other elements/mechanics**.

For example, in this level, the middle column would need a minimum of 4 vertical booster creation - costing the player approximately 8 moves - which in-turn might increase the difficulty of level completion.





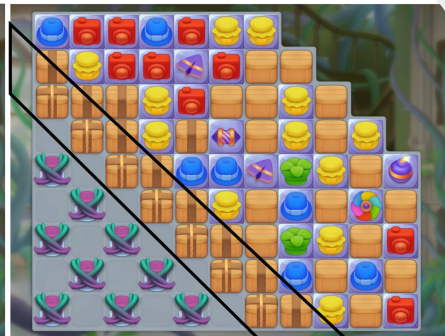
Difficulty creation:

Grid design and visual difficulty

Visual distinction: Use of empty cells provide clear visual guidance on priority areas.

Visual clutter?: The use of “Wooden Boxes” and yellow coloured tiles creates a visual clutter that might affect a player’s reading of the grid.

Difficulty in reading key structural arrangements (like the diamond pattern of boxes) might reduce the player's pattern recognition speed. Consequently, this could slow down the rate of initial turn planning and lead to a less efficient start to the level.





Difficulty creation: Grass mechanic

The 'Grass' mechanic in Merge Gardens shares the fundamental goal with the 'Jelly' mechanic in Match Villains. However, Merge gardens offer better player experience:

- **Visual clarity:** The element's design is highly contrasting and visually striking for immediate player recognition.
- **Strategic Grid Placement:** Unlike the comparative mechanic, Grass is systematically placed in low-element density areas. This placement strategy eliminates visual noise, allowing players to quickly identify the objective and formulate their optimal move sequence.



Thank you for reading!
Supriya Dutta

